

Spring Boot Enterprise Error Diagnostics & Resolution Guide

Purpose: Comprehensive Handbook & Knowledge Base for LogMonito AI / RAG Diagnostic Integration

Scope: Spring Boot 3.x / Java 21 • Microservices • Kafka • MongoDB • Resilience • Security

0. Global Exception Handling Pattern (Spring Boot 3 / RFC 9457)

Modern Spring Boot applications standardize error responses using RFC 9457 `ProblemDetail` managed by a central `@RestControllerAdvice`:

```
@RestControllerAdvice
public class GlobalExceptionHandler extends ResponseEntityExceptionHandler {

    @ExceptionHandler(Exception.class)
    public ProblemDetail handleUnhandled(Exception ex) {
        ProblemDetail problem = ProblemDetail.forStatusAndDetail(
            HttpStatus.INTERNAL_SERVER_ERROR, "An unexpected error occurred.");
        problem.setTitle("Internal Server Error");
        problem.setProperty("timestamp", Instant.now());
        return problem;
    }

    @Override
    protected ResponseEntity<Object> handleMethodArgumentNotValid(
        MethodArgumentNotValidException ex, HttpHeaders headers, HttpStatusCode status, WebRequest request) {
        ProblemDetail problem = ProblemDetail.forStatusAndDetail(
            HttpStatus.BAD_REQUEST, "Validation constraints failed.");
        Map<String, String> errors = new HashMap<>();
        ex.getBindingResult().getFieldErrors().forEach(e -> errors.put(e.getField(), e.getDefaultMessage()));
        problem.setProperty("fieldErrors", errors);
        return ResponseEntity.status(status).body(problem);
    }
}
```

1. Startup & Dependency Injection Errors

ERR-START-001: NoSuchBeanDefinitionException

FATAL

Category: Spring IoC • **Symptom:** Application startup crash

```
org.springframework.beans.factory.NoSuchBeanDefinitionException: No qualifying bean of type 'com.ex
```

Root Cause: Spring failed to locate a bean of the requested type for constructor/field injection.

- **Remediation 1:** Ensure the class is annotated with `@Service`, `@Component`, or `@Repository`.
- **Remediation 2:** Check package scanning. Ensure the class resides under the main `@SpringBootApplication` package or add `@ComponentScan`.
- **Remediation 3:** If conditionally loaded, check if the required property (e.g., `@ConditionalOnProperty`) is enabled in `application.yml`.

ERR-START-002: BeanCurrentlyInCreationException (Circular Dependency)

FATAL

Category: Spring Lifecycle • **Symptom:** Context fails with circular dependency graph

```
The dependencies of some of the beans in the application context form a cycle:  
serviceA -> serviceB -> serviceA
```

Root Cause: Two or more beans mutually inject each other at instantiation time. Spring Boot 2.6+ strictly forbids this by default.

- **Remediation 1 (Best Practice):** Refactor by extracting the mutually shared logic into a separate third service or using Spring Application Events.
- **Remediation 2:** Break the eager construction cycle using `@Lazy` on one of the constructor parameters:

```
public ServiceA(@Lazy ServiceB b) {
```

ERR-START-003: PortInUseException

FATAL

Category: Embedded Web Server • **Symptom:** Web server failed to start

```
org.springframework.boot.web.server.PortInUseException: Port 8080 was already in use.
```

Root Cause: Another application or previous zombie instance has already bound port 8080.

- **Remediation 1:** Configure a distinct port in `application.properties`: `server.port=8085`.
- **Remediation 2:** Terminate the offending PID: Windows `netstat -ano | findstr :8080` then `taskkill /PID <PID> /F`; Linux `fuser -k 8080/tcp`.

2. Web & Request Handling Errors

ERR-WEB-001: MethodArgumentNotValidException

HTTP 400

Category: REST / Jakarta Validation • **Symptom:** Client gets 400 Bad Request

```
MethodArgumentNotValidException: Validation failed for argument [0]: Field error in 'userDto' on fi
```

Root Cause: An incoming JSON request payload violated one or more DTO annotations (e.g. `@NotBlank`, `@Email`, `@Min`).

- **Remediation:** Intercept via `handleMethodArgumentNotValid` in `@RestControllerAdvice`, extract all field violations into a structured error dictionary, and return RFC 9457 `ProblemDetail`.

ERR-WEB-002: HttpMessageNotReadableException

HTTP 400

Category: JSON Deserialization • **Symptom:** Unparseable JSON body

```
HttpMessageNotReadableException: JSON parse error: Cannot deserialize value of type Instant from St
```

Root Cause: Jackson JSON parsing failure due to invalid syntax, non-matching date/time formats, or unknown Enum values.

- **Remediation:** Add `@JsonFormat(pattern = "yyyy-MM-dd'T'HH:mm:ss.SSSX")` on date fields, or configure `DeserializationFeature.READ_UNKNOWN_ENUM_VALUES_AS_NULL`.

3. Database & Persistence Errors

ERR-DB-001: Connection Pool Exhaustion (HikariPool Timeout)

HTTP 500 / 503

Category: Database / Connection Pool • **Symptom:** Cascading API latency & timeouts

```
java.sql.SQLTransientConnectionException: HikariPool-1 - Connection is not available, request timed
```

Root Cause: All connections in the pool are checked out. Threads are blocked waiting for connections to be returned.

- **Remediation 1:** Ensure long external REST/Kafka calls are **never** executed inside a `@Transactional` method.
- **Remediation 2:** Enable connection leak detection: `spring.datasource.hikari.leak-detection-threshold=2000`.
- **Remediation 3:** Tune pool size: `spring.datasource.hikari.maximum-pool-size=20` and add indexes to slow queries.

ERR-DB-002: MongoSocketOpenException / Connection Refused

DATABASE DOWN

Category: NoSQL / MongoDB • **Symptom:** Storage/Auth operations fail completely

```
com.mongodb.MongoSocketOpenException: Exception opening socket
Caused by: java.net.ConnectException: Connection refused: no further information
```

Root Cause: MongoDB daemon is down, not bound to localhost/0.0.0.0, or container port 27017 is unreachable.

- **Remediation:** Verify MongoDB container is running with `docker ps` and verify `spring.data.mongodb.uri` host/credentials.

ERR-DB-003: LazyInitializationException

HTTP 500

Category: JPA / Hibernate • **Symptom:** Failure during entity serialization

```
org.hibernate.LazyInitializationException: could not initialize proxy [com.example.Order#1] - no Se
```

Root Cause: Accessing a `fetch = FetchType.LAZY` child property after the Hibernate session / transaction has already closed.

- **Remediation (Best):** Use DTO projections and join-fetch queries (`SELECT o FROM Order o JOIN FETCH o.items`) or `@EntityGraph` .

4. Event-Driven Messaging Errors (Apache Kafka)

ERR-MSG-001: RecordDeserializationException / ClassNotFoundException

CONSUMER BLOCKED

Category: Apache Kafka / Deserialization • **Symptom:** Listener loop continuously fails

```
ListenerExecutionFailedException: Error deserializing key/value for partition raw-logs-topic-0
Caused by: java.lang.ClassNotFoundException: com.example.IngestionService.dto.LogPayload
```

Root Cause: Kafka `JsonSerializer` injected producer class headers, but the consumer lives in another package or module.

- **Remediation 1:** Configure `spring.kafka.consumer.properties.spring.json.type.mapping` to map producer DTO to consumer DTO.
- **Remediation 2:** Wrap value deserializer with Spring's `ErrorHandlingDeserializer` and route poison-pill messages to a Dead-Letter Topic (DLT).

5. JVM, Memory & Runtime Errors

ERR-JVM-001: OutOfMemoryError: Java heap space

JVM CRASH

Category: JVM Resource • **Symptom:** Application becomes unresponsive and terminates

```
java.lang.OutOfMemoryError: Java heap space
Dumping heap to java_pid.hprof ...
```

Root Cause: The application allocated more memory than available in `-Xmx`, and garbage collection reclaimed insufficient memory.

- **Remediation 1:** Enforce database pagination: never allow unrestricted `findAll()` across unbounded tables.
- **Remediation 2:** Capture heap dump on crash: `-XX:+HeapDumpOnOutOfMemoryError -XX:HeapDumpPath=/dumps` and inspect with Eclipse MAT.

6. Summary Diagnostic Matrix for AI / RAG Systems

Code	Exception Class	Status	Primary Root Cause	First Diagnostic Action
ERR-START-001	<code>NoSuchBeanDefinitionException</code>	Startup Crash	Missing bean annotation or package scan mismatch	Check <code>@Service</code> and base package
ERR-START-002	<code>BeanCurrentlyInCreationException</code>	Startup Crash	Circular dependency between beans	Refactor or apply <code>@Lazy</code>
ERR-WEB-001	<code>MethodArgumentNotValidException</code>	HTTP 400	Jakarta <code>@Valid</code> constraints failed	Inspect field errors in <code>ProblemDetail</code>
ERR-DB-001	<code>CannotGetJdbcConnectionException</code>	HTTP 500/503	Hikari connection pool exhaustion	Check slow queries & non-DB code in transactions
ERR-DB-002	<code>MongoSocketOpenException</code>	Database Down	MongoDB unreachable or port not open	Verify MongoDB container & connection URI
ERR-MSG-001	<code>RecordDeserializationException</code>	Consumer Loop Error	Kafka type-header mismatch or poison message	Configure type mapping or <code>ErrorHandlingDeserializer</code>
ERR-JVM-	<code>OutOfMemoryError</code>	JVM Terminated	Heap exhaustion / unbounded	Inspect heap dump & enforce query pagination

Code	Exception Class	Status	Primary Root Cause	First Diagnostic Action
001			memory allocation	